# CSE 101 - Local Development with Docker

This guide helps you run your C programs in a **Linux environment** that matches the one used for grading.

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## Why use Docker?

Your code will be graded in a Linux environment with:

- **Ubuntu 22.04**
- **gcc (C compiler)**
- **make**
- **valgrind** (for memory checks)

Using Docker lets you run your code locally under the **same environment**, so you avoid "works on my machine" issues.

If you choose not to use Docker, you are responsible for making sure your environment matches this setup.

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## 1. Install Docker

Install Docker on your machine:

- Official guide: https://docs.docker.com/get-docker/
- Windows / macOS: install **Docker Desktop**
- Linux: install **Docker Engine**

Verify installation:

```
docker --version
docker run --rm hello-world
```

## 2. Set up your project

Place the provided Dockerfile inside your assignment folder.

Example:

```
hw0/
    main.c
    Makefile
    Dockerfile
```

## 3. Build the environment (one-time)

From inside your assignment folder:

```
docker build -t cse101-build .
```

This creates a local Linux environment with the required tools.

## 4. Compile your code

Run make inside the container:

macOS / Linux

```
docker run --rm -v "${PWD}:/workspace" -w /workspace cse101-build make
```

Windows (PowerShell)

```
docker run --rm -v "${PWD}:/workspace" -w /workspace cse101-build make
```

## 5. Run your program

After compiling:

```
docker run --rm -v "${PWD}:/workspace" -w /workspace cse101-build ./main
```

Your program name depends on your Makefile. Replace `./main` with your executable name if different.

## 6. (Optional) Check memory with valgrind

```
docker run --rm -v "${PWD}:/workspace" \
-w /workspace cse101-build valgrind --leak-check=full ./main
```